Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “recoil” and “external” to explain your model or result. Sentence starters are optional.

recoil: motion backward when another part is sent forward

external: acting from outside the chosen system

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A 3 kg cart travels left at 5 m/s.
Find momentum and draw its direction.

Word help: momentum = mass times velocity, including direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part’s momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two carts push apart from rest on an ideal track. **Draw** their momentum arrows. How must the two momenta compare?

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



A moves left B moves right

Marzano check - circle: 1 2 3

04 Apply the physics

A 1 kg cart moves left at 4 m/s after release. A 2 kg cart moves right. To keep total momentum zero, find the second cart’s speed. Organize the known quantities on the before-and-after model.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two balls launch horizontally from the same height at 4 and 24 m/s. Ignore air resistance.
Compare fall times and horizontal ranges.
Label separate horizontal and vertical motion.

Word help: horizontal = along a level line, usually the x direction

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg cart leaves with momentum 6 kg·m/s left. What must the other cart’s momentum be if total was zero? Organize the known quantities on the before-and-after model.

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep a 1 kg cart's departure velocity +6 m/s. **Choose** a second cart mass that recoils at -2 m/s. Organize the known quantities on the before-and-after model.

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3